



Length, area and scale

Teacher resource

Question 1

Bill makes scale models of old ships. The length of one scale model is 118 centimetres. The length of the actual ship was 59 metres.

Which of the following represents the scale Bill used to make his model?

1:2

☐

1:50

☐

59:118

☐

118:59

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Answer: 1:50

Skill: Students represent a scale as a ratio.

Background information

Students should understand that the shape of a two-dimensional figure or three-dimensional object does not change when it is reduced or enlarged. 'Same shape' in this sense means that all angles are maintained and the lengths of sides and diagonals all change by the same multiple.

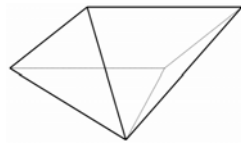
By contrast, some students may interpret the phrase 'same shape' to mean 'category' of figures or objects, such as triangles or pyramids. Within a category such as triangles, however, any two triangles may or may not be considered to have the same 'shape' when the angles and ratios of sides are considered.

Additional questions

1. A sail on the original ship had a length of 20 metres and width of 5 metres. What are the dimensions of this sail on the scale model?
2. What area on the scale model would represent one square metre on the actual ship?
3. What area on the actual ship would be represented by one square millimetre on the model?

Question 2

A jeweller cuts a gemstone so that the top face is a square. The length of the square is 8mm.



The jeweller constructs a scale drawing of the top face. The length of the square which represents the top face is 4cm.

Which of the following is the scale used in this drawing?

- ☐ 1 cm represents 5 mm
- ☐ 1 cm represents 2mm
- ☐ 2 cm represents 1mm
- ☐ 5 cm represents 1mm

Answer: B

Skill: Students represent a scale as a ratio.

Additional questions

1. The other four faces of the gemstone are isosceles triangles, with the equal sides measuring 15mm. A scale drawing of one of these triangles is constructed using the same scale. What is the length of the equal sides in the scale drawing?
2. How would the angles in the scale drawing of the triangular faces compare to the angles in the gemstone?

Question 3

Andrea wants to enlarge a photo which measures 24cm by 15cm. She would like to transfer the enlarged photo to a canvas which measures 1.25m by 0.9m. Andrea would like to have the enlarged photo as large as possible without distorting its shape.

What scale factor should she use to enlarge the photo?

Answer: 5.2

Skill: Students interpret similarity in a context.

Additional questions

1. What are the dimensions of the enlarged photo?
2. Andrea also wanted to reduce the photo to fit on the lid of a box which measures 16 cm by 12 cm. By what scale factor should the photo be reduced so that it fits on the lid of the box and its shape is not distorted?

Question 4

A triangle, with sides 3 cm, 4 cm and 5 cm, is enlarged or reduced. Which triangles could not be produced?

- ☐ A triangle with sides 0.75 cm, 1 cm and 1.25 cm.
- ☐ A triangle with sides 1 cm, 2 cm and 3 cm.
- ☐ A triangle with sides 1.5 cm, 2 cm and 2.5 cm.
- ☐ A triangle with sides 4 cm, 5 cm and 6 cm.

Answer: B and D

Skill: Students identify similar triangles.

Additional questions

1. How does the area of 'triangle C' compare to the area of the original triangle?
2. How does the area of 'triangle A' compare to the area of the original triangle?

Student activity: length, area and scale

Question 1

Bill makes scale models of old ships. His scale model of one ship was 118 centimetres long. The real ship was 59 metres long.

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Question 2

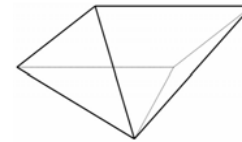
Andrea wants to enlarge a photo which measures 24cm by 15cm. She would like to transfer the enlarged photo to a canvas which measures 1.25m by 0.9m.

Andrea would like to have the enlarged photo as large as possible without distorting its shape.

What scale factor should she use to enlarge the photo?

Question 3

A jeweller cut a gemstone so that the top surface is a square with sides that are 8mm long.



The jeweller constructed a scale drawing of this top surface. He drew a square with 4 cm sides. Which of the following is the scale used in this drawing?

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1 cm represents 5 mm

☐

1 cm represents 2mm

☐

2 cm represents 1mm

☐

5 cm represents 1mm

Question 4

A triangle, with sides 3 cm, 4 cm and 5 cm, is enlarged or reduced.

Which triangles could not be produced?

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A triangle with sides 0.75 cm, 1 cm and 1.25 cm.

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A triangle with sides 1 cm, 2 cm and 3 cm.

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A triangle with sides 1.5 cm, 2 cm and 2.5 cm.

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A triangle with sides 4 cm, 5 cm and 6 cm.